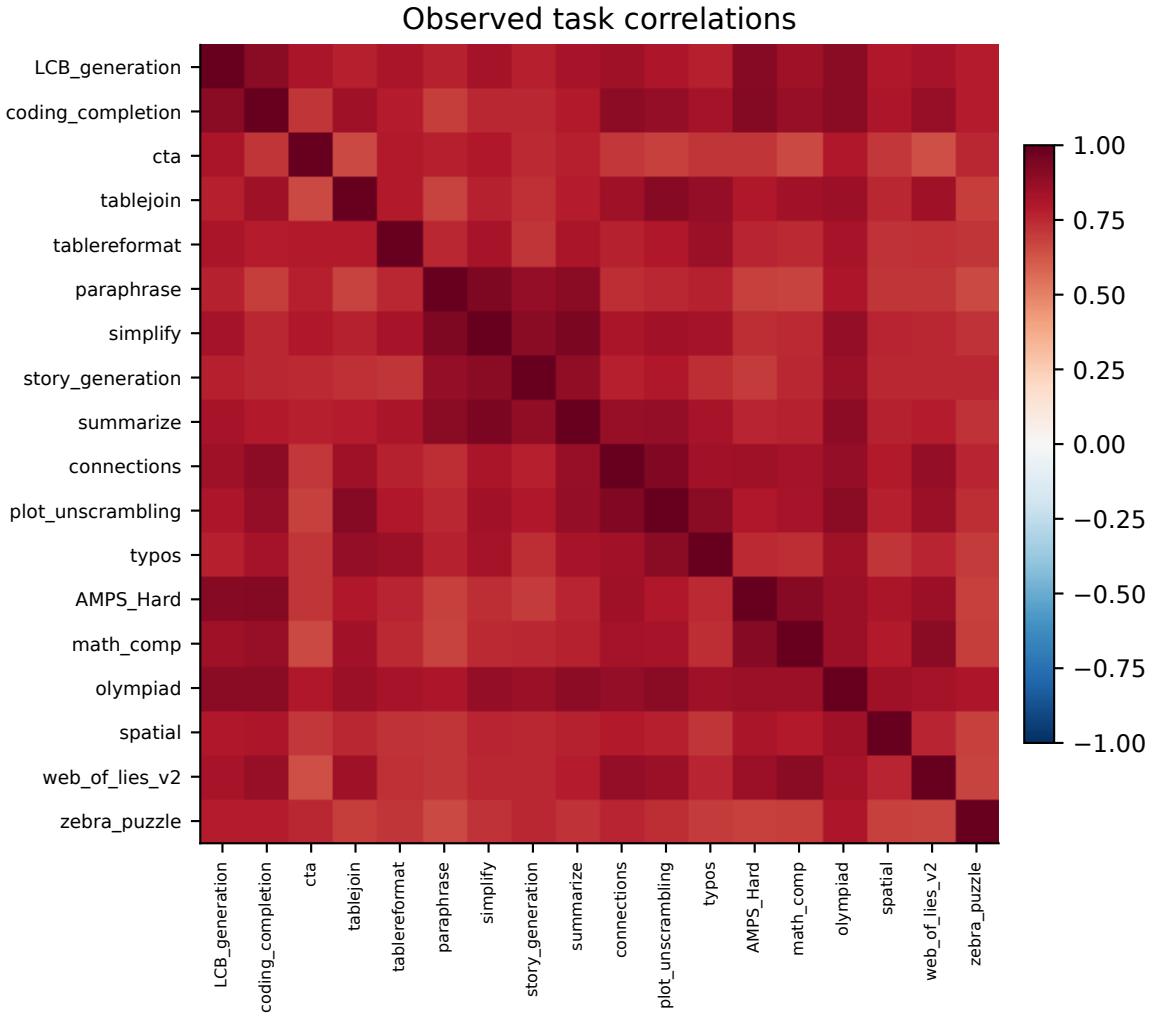
A strong shared factor; residual structure remains

65 complete models | 18 tasks | 6 domains | 1,000 common question IDs



One-factor g accounts for 79.6% of standardized task variance. Family/item interval: 79.6% [68.3, 80.2].

One factor: CFI 0.817; RMSEA 0.204; residual RMS 0.050. High shared variance does not establish adequate one-factor fit.

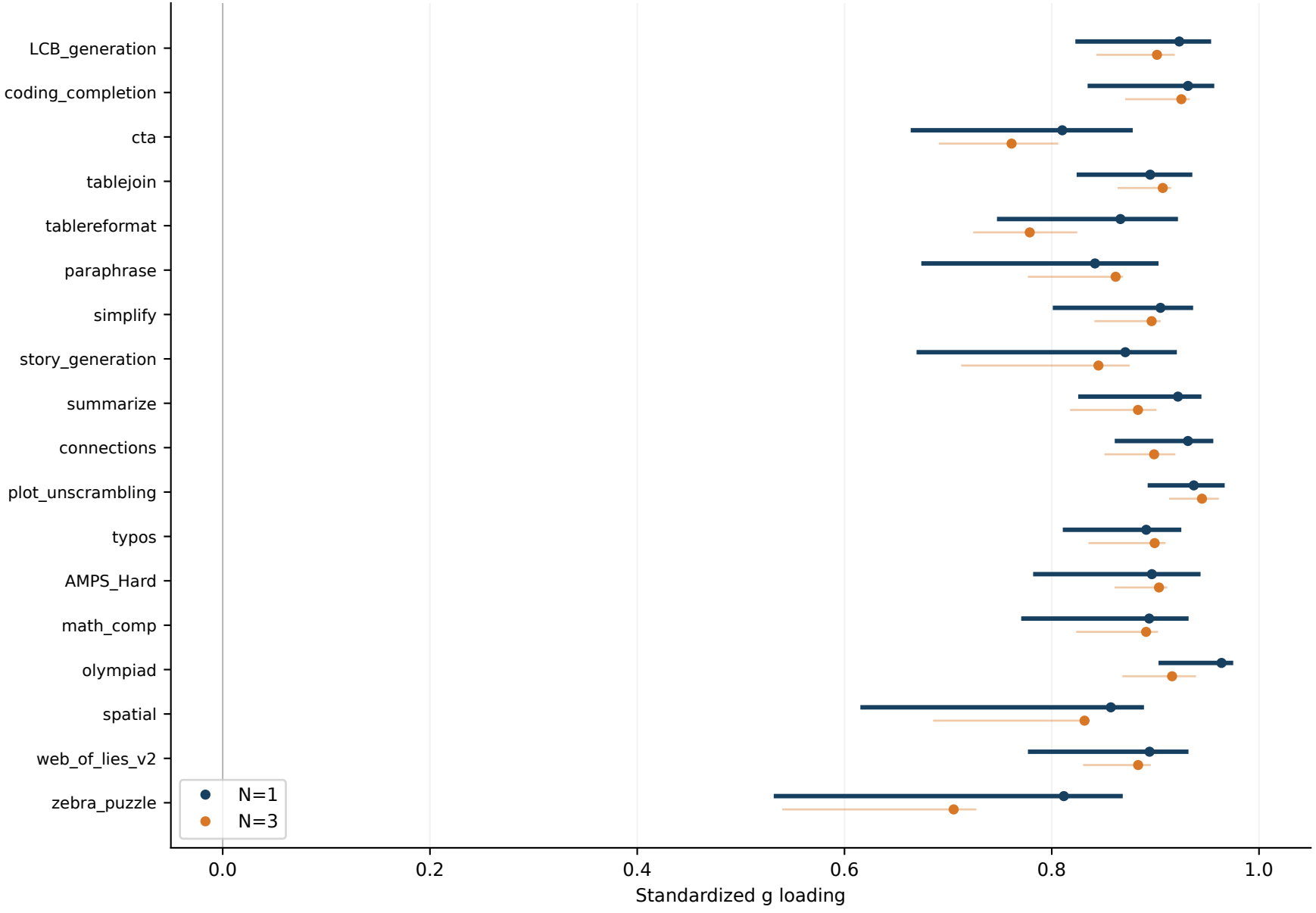
Three-group hierarchy: CFI 0.898; RMSEA 0.155. BIC difference (hierarchy - one factor): -150.1.

Hierarchical g accounts for 75.8% of task variance. Unit-composite omega_h is 0.949; these are different quantities.

These are statistical performance dimensions within a selected historical sample. They are neither human IQ nor evidence of an increase or decline in generality over time.

General-factor task loadings

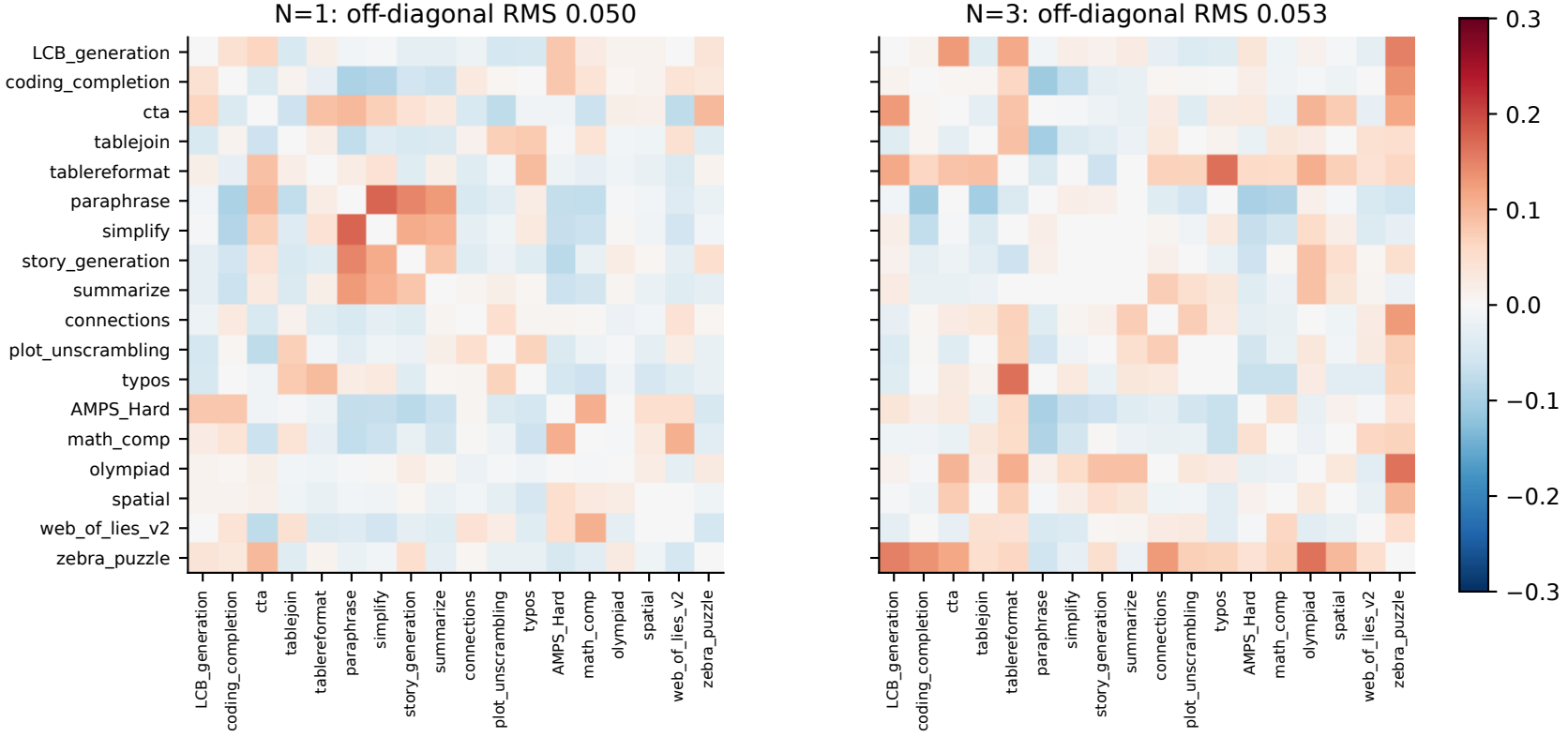
Standardized loadings | thick intervals: family + paired items | thin intervals: paired items only



Intervals are withheld when fewer than 90% of bootstrap fits are admissible. No task was excluded for a weak or negative loading.
Some item-only ranges exclude their point estimate; finite-item resampling bias is material. Coverage is not calibrated; see bias_diagnostics.csv.

Residual correlations expose what g misses

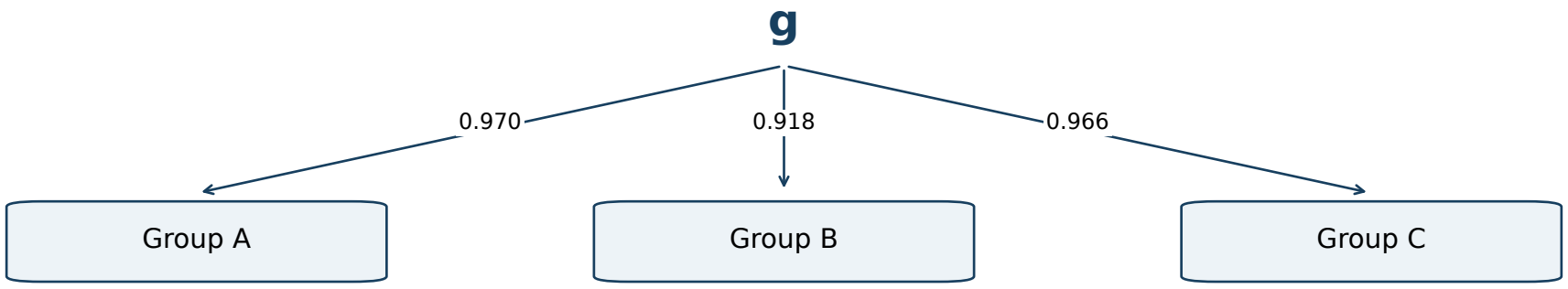
Observed correlation minus fitted correlation | diagonal omitted | identical color scales



Task grouping is exploratory and uses the same sample. Better residual fit alone does not independently validate the hierarchy.

The three-group hierarchy

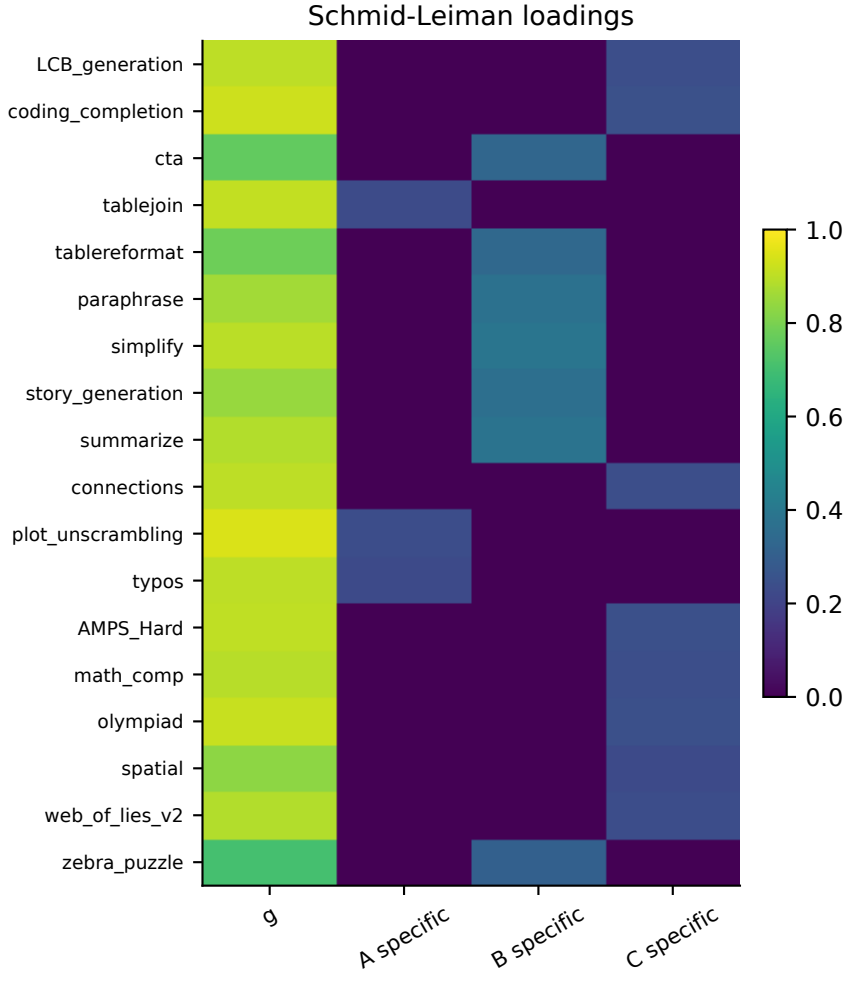
Existing PAF/varimax assignment and ML higher-order model | group names are neutral



Group A (3 tasks):
- tablejoin
- plot_unscrambling
- typos

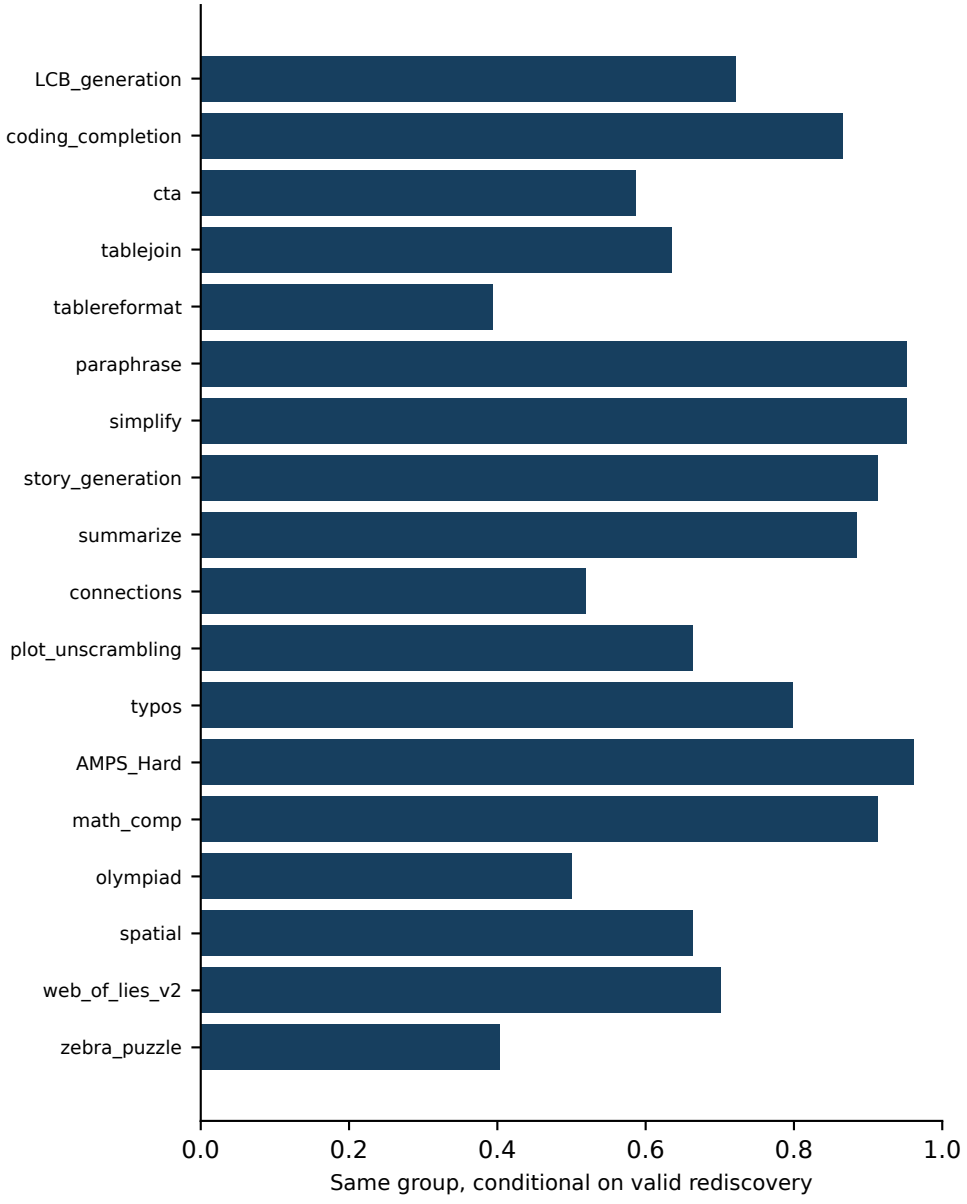
Group B (7 tasks):
- cta
- tablereformat
- paraphrase
- simplify
- story_generation
- summarize
- zebra_puzzle

Group C (8 tasks):
- LCB_generation
- coding_completion
- connections
- AMPS_Hard
- math_comp
- olympiad
- spatial
- web_of_lies_v2



Bootstrap validity and group stability

1000 replicates per interval mode | 200 rediscovery replicates | seed 0



46 family clusters; 39 unresolved model identities remain singleton clusters. This is a dependence sensitivity, not proof that clusters are independent.

item, N=1: 1000/1000 admissible; 0 boundary fits. Intervals reported.

item, N=3: 996/1000 admissible; 4 boundary fits. Intervals reported.

family_item, N=1: 1000/1000 admissible; 0 boundary fits. Intervals reported.

family_item, N=3: 782/1000 admissible; 218 boundary fits. Intervals withheld.

Rediscovery: 104/200 paired admissible comparisons; hierarchy has lower BIC in 104. Failed/nonviable hierarchies are not evidence against domain structure.

Fixed-group intervals condition on the discovered assignment. Group-stability bars are conditional on admissible rediscovery and align group labels before comparison.

Relative model scores and ranks (1-23)

Family + paired-item intervals | positive rank shift = lower rank under the hierarchy

Model	g1	95% CI	Rank [95%]	g3	95% CI	Rank [95%]	Rank shift [95%]
claude-3-5-sonnet-20240620	1.88	[1.32, 2.37]	1 [1, 9]	2.33	withheld	1 [withheld]	0 [withheld]
gpt-4o-2024-08-06	1.71	[1.14, 2.16]	2 [1, 12]	1.67	withheld	3 [withheld]	1 [withheld]
chatgpt-4o-latest	1.56	[1.01, 1.95]	3 [1, 13]	1.48	withheld	6 [withheld]	3 [withheld]
gpt-4o-2024-05-13	1.53	[0.99, 1.93]	4 [1, 14]	1.31	withheld	7 [withheld]	3 [withheld]
Meta-Llama-3.1-405B-Instruct-Turbo	1.53	[0.98, 2.01]	5 [1, 14]	1.66	withheld	4 [withheld]	-1 [withheld]
gpt-4-turbo-2024-04-09	1.41	[0.77, 1.88]	6 [2, 18]	1.28	withheld	8 [withheld]	2 [withheld]
gemini-1.5-pro-exp-0801	1.37	[0.68, 1.84]	7 [1, 19]	1.13	withheld	11 [withheld]	4 [withheld]
gpt-4-1106-preview	1.35	[0.81, 1.79]	8 [2, 17]	1.16	withheld	9 [withheld]	1 [withheld]
claude-3-opus-20240229	1.29	[0.77, 1.88]	9 [2, 17]	1.94	withheld	2 [withheld]	-7 [withheld]
Meta-Llama-3.1-70B-Instruct-Turbo	1.18	[0.55, 1.70]	10 [2, 22]	1.05	withheld	15 [withheld]	5 [withheld]
coding-meta-llama-3.1-70b-instruct-chk-50	1.15	[0.59, 1.56]	11 [4, 20]	1.14	withheld	10 [withheld]	-1 [withheld]
gpt-4-0125-preview	1.13	[0.54, 1.64]	12 [3, 22]	0.77	withheld	19 [withheld]	7 [withheld]
mistral-large-2407	1.11	[0.48, 1.66]	13 [4, 23]	0.97	withheld	16 [withheld]	3 [withheld]
deepseek-coder	1.03	[0.41, 1.49]	14 [5, 24]	1.13	withheld	12 [withheld]	-2 [withheld]
gpt-4-0613	1.00	[0.47, 1.57]	15 [4, 22]	1.07	withheld	14 [withheld]	-1 [withheld]
deepseek-chat	0.99	[0.44, 1.47]	16 [5, 24]	0.95	withheld	17 [withheld]	1 [withheld]
gemini-1.5-pro-latest	0.86	[0.39, 1.42]	17 [6, 25]	1.49	withheld	5 [withheld]	-12 [withheld]
gpt-4o-mini-2024-07-18	0.74	[0.12, 1.19]	18 [10, 28]	0.74	withheld	20 [withheld]	2 [withheld]
lcb-math-qwen2-72b-instructv3-merged-50	0.72	[0.15, 1.18]	19 [10, 29]	1.09	withheld	13 [withheld]	-6 [withheld]
Qwen2-72B-Instruct	0.69	[0.09, 1.22]	20 [10, 30]	0.77	withheld	18 [withheld]	-2 [withheld]
gemma-2-27b-it	0.61	[-0.06, 1.13]	21 [11, 32]	0.39	withheld	23 [withheld]	2 [withheld]
gemini-1.5-flash-latest	0.58	[0.07, 1.18]	22 [11, 30]	0.73	withheld	21 [withheld]	-1 [withheld]
mistral-large-2402	0.50	[-0.12, 1.03]	23 [13, 33]	-0.09	withheld	29 [withheld]	6 [withheld]

Scores are centered and scaled within the original 65-model panel in each replicate. They have no absolute or human-IQ interpretation.

Rank intervals are marginal; overlapping intervals are not a pairwise significance test. A point rank is not a precise ordering.

Relative model scores and ranks (24-46)

Family + paired-item intervals | positive rank shift = lower rank under the hierarchy

Model	g1	95% CI	Rank [95%]	g3	95% CI	Rank [95%]	Rank shift [95%]
claude-3-sonnet-20240229	0.46	[-0.01, 0.95]	24 [14, 31]	0.22	withheld	24 [withheld]	0 [withheld]
Meta-Llama-3-70B-Instruct	0.44	[-0.09, 1.08]	25 [12, 34]	-0.07	withheld	28 [withheld]	3 [withheld]
claude-3-haiku-20240307	0.22	[-0.35, 0.80]	26 [17, 40]	-0.31	withheld	36 [withheld]	10 [withheld]
qwen2-math-72b-instruct	0.17	[-0.46, 0.73]	27 [18, 42]	0.63	withheld	22 [withheld]	-5 [withheld]
command-r-plus	0.12	[-0.50, 0.71]	28 [19, 43]	-0.06	withheld	27 [withheld]	-1 [withheld]
gpt-3.5-turbo-0125	0.09	[-0.54, 0.59]	29 [21, 44]	-0.11	withheld	30 [withheld]	1 [withheld]
gpt-3.5-turbo-1106	0.07	[-0.42, 0.59]	30 [21, 42]	-0.29	withheld	35 [withheld]	5 [withheld]
mistral-small-2402	0.05	[-0.45, 0.56]	31 [21, 42]	-0.17	withheld	33 [withheld]	2 [withheld]
gemma-2-9b-it	-0.11	[-0.64, 0.41]	32 [23, 46]	0.14	withheld	25 [withheld]	-7 [withheld]
Phi-3-medium-4k-instruct	-0.14	[-1.00, 0.49]	33 [22, 54]	-0.42	withheld	38 [withheld]	5 [withheld]
Qwen1.5-110B-Chat	-0.17	[-0.76, 0.33]	34 [26, 49]	-0.55	withheld	42 [withheld]	8 [withheld]
Qwen1.5-72B-Chat	-0.17	[-0.75, 0.40]	35 [25, 49]	-0.52	withheld	40 [withheld]	5 [withheld]
Phi-3-medium-128k-instruct	-0.25	[-1.00, 0.32]	36 [25, 54]	-0.36	withheld	37 [withheld]	1 [withheld]
open-mistral-nemo	-0.28	[-0.82, 0.24]	37 [27, 50]	-0.17	withheld	32 [withheld]	-5 [withheld]
DeepSeek-Coder-V2-Lite-Instruct	-0.28	[-0.98, 0.23]	38 [27, 54]	-0.25	withheld	34 [withheld]	-4 [withheld]
Meta-Llama-3.1-8B-Instruct-Turbo	-0.31	[-0.76, 0.25]	39 [27, 50]	-0.16	withheld	31 [withheld]	-8 [withheld]
Meta-Llama-3-8B-Instruct	-0.32	[-0.78, 0.19]	40 [27, 49]	-0.64	withheld	46 [withheld]	6 [withheld]
Qwen2-7B-Instruct	-0.44	[-0.97, 0.07]	41 [30, 53]	-0.42	withheld	39 [withheld]	-2 [withheld]
Phi-3-small-8k-instruct	-0.44	[-0.94, 0.04]	42 [30, 53]	-0.68	withheld	48 [withheld]	6 [withheld]
command-r	-0.44	[-0.95, 0.08]	43 [30, 53]	-0.77	withheld	54 [withheld]	11 [withheld]
Phi-3-small-128k-instruct	-0.50	[-1.16, 0.10]	44 [29, 58]	-1.40	withheld	61 [withheld]	17 [withheld]
OpenHermes-2.5-Mistral-7B	-0.55	[-1.05, -0.07]	45 [33, 55]	-0.66	withheld	47 [withheld]	2 [withheld]
mathstral-7B-v0.1	-0.65	[-1.22, -0.09]	46 [34, 58]	0.11	withheld	26 [withheld]	-20 [withheld]

Scores are centered and scaled within the original 65-model panel in each replicate. They have no absolute or human-IQ interpretation.

Rank intervals are marginal; overlapping intervals are not a pairwise significance test. A point rank is not a precise ordering.

Relative model scores and ranks (47-65)

Family + paired-item intervals | positive rank shift = lower rank under the hierarchy

Model	g1	95% CI	Rank [95%]	g3	95% CI	Rank [95%]	Rank shift [95%]
Mistral-7B-Instruct-v0.3	-0.71	[-1.15, -0.15]	47 [33, 57]	-0.52	withheld	41 [withheld]	-6 [withheld]
Phi-3-mini-128k-instruct	-0.73	[-1.29, -0.17]	48 [35, 59]	-0.93	withheld	56 [withheld]	8 [withheld]
Mistral-7B-Instruct-v0.2	-0.74	[-1.23, -0.16]	49 [33, 59]	-0.76	withheld	53 [withheld]	4 [withheld]
Phi-3-mini-4k-instruct	-0.84	[-1.32, -0.24]	50 [36, 60]	-0.63	withheld	44 [withheld]	-6 [withheld]
gemma-1.1-7b-it	-0.88	[-1.27, -0.35]	51 [39, 59]	-0.74	withheld	50 [withheld]	-1 [withheld]
zephyr-7b-alpha	-0.94	[-1.40, -0.34]	52 [38, 60]	-0.57	withheld	43 [withheld]	-9 [withheld]
Qwen1.5-7B-Chat	-0.95	[-1.38, -0.45]	53 [42, 61]	-0.75	withheld	52 [withheld]	-1 [withheld]
Starling-LM-7B-beta	-1.01	[-1.58, -0.30]	54 [37, 63]	-0.74	withheld	49 [withheld]	-5 [withheld]
DeepSeek-V2-Lite-Chat	-1.03	[-1.45, -0.51]	55 [42, 61]	-0.85	withheld	55 [withheld]	0 [withheld]
zephyr-7b-beta	-1.08	[-1.49, -0.53]	56 [42, 62]	-0.64	withheld	45 [withheld]	-11 [withheld]
vicuna-7b-v1.5-16k	-1.22	[-1.63, -0.78]	57 [48, 64]	-0.99	withheld	58 [withheld]	1 [withheld]
vicuna-7b-v1.5	-1.33	[-1.73, -0.79]	58 [49, 64]	-0.94	withheld	57 [withheld]	-1 [withheld]
Llama-2-7b-chat-hf	-1.35	[-1.92, -0.79]	59 [49, 65]	-0.74	withheld	51 [withheld]	-8 [withheld]
Qwen1.5-4B-Chat	-1.45	[-1.88, -0.80]	60 [50, 65]	-1.40	withheld	60 [withheld]	0 [withheld]
Yi-6B-Chat	-1.50	[-1.92, -0.89]	61 [51, 65]	-1.94	withheld	65 [withheld]	4 [withheld]
Qwen2-1.5B-Instruct	-1.50	[-1.85, -0.94]	62 [52, 65]	-1.56	withheld	62 [withheld]	0 [withheld]
Qwen2-0.5B-Instruct	-1.66	[-1.94, -1.05]	63 [55, 65]	-1.15	withheld	59 [withheld]	-4 [withheld]
Qwen1.5-1.8B-Chat	-1.79	[-2.12, -1.08]	64 [55, 65]	-1.57	withheld	63 [withheld]	-1 [withheld]
Qwen1.5-0.5B-Chat	-1.80	[-2.12, -1.12]	65 [56, 65]	-1.83	withheld	64 [withheld]	-1 [withheld]

Scores are centered and scaled within the original 65-model panel in each replicate. They have no absolute or human-IQ interpretation.

Rank intervals are marginal; overlapping intervals are not a pairwise significance test. A point rank is not a precise ordering.

Sensitivity and matched historical comparisons

Changing cohorts and changing batteries must not be mistaken for evolution in gAI

Archive	Models	Tasks	g variance	CFI	RMSEA	Panel
2024-06-26	50	17	78.9%	0.804	0.217	full-battery
2024-06-26	50	10	77.5%	0.866	0.228	coverage-selected
2024-07-20	57	17	78.1%	0.802	0.215	full-battery
2024-07-20	57	11	76.4%	0.839	0.234	coverage-selected
2024-07-29	59	18	78.8%	0.812	0.205	full-battery
2024-07-29	59	11	78.1%	0.858	0.224	coverage-selected
2024-08-04	60	18	79.1%	0.819	0.201	full-battery
2024-08-04	60	12	79.2%	0.849	0.224	coverage-selected
2024-08-18	65	18	79.6%	0.817	0.204	full-battery
2024-08-18	65	13	79.7%	0.821	0.239	coverage-selected
2024-10-22	46	18	60.7%	0.702	0.212	full-battery
2024-10-22	84	11	76.0%	0.841	0.238	coverage-selected

Matched 86ad8e20 -> 65fce612: 59 common models, 18 tasks, 1000 common question IDs; 0 changed item scores. Maximum task-mean change 0.0000; maximum g-loading change 0.0000.

Matched 65fce612 -> 81f6a1cd: 60 common models, 18 tasks, 1000 common question IDs; 17 changed item scores. Maximum task-mean change 0.1000; maximum g-loading change 0.0237.

The matched comparisons hold the observed identities and battery fixed. Historical API versions/settings remain incompletely verified, so even changed item scores do not by themselves demonstrate model improvement.

Methods, provenance and practical limits

All inputs are public, pinned local files; this analysis makes no inference/API requests

Input score revision: 81f6a1cd42f793285e215aabed4fd1c71a94889a; battery: 2024-07-26. Exact model IDs are retained. Six incomplete models are excluded from the primary matrix; exclusions and item IDs are exported.

One-factor and three-group covariance models are fitted by ML with analytic gradients and multiple starts. Group discovery uses PAF/varimax. Factor variances are fixed to one; signs are aligned positive. Hierarchies require three groups with at least three indicators each. Two-group hierarchical g is not identified and is excluded.

The one-factor standardized loading-square mean is g-attributed task variance. Hierarchical ECV is g variance divided by total common variance. Omega_h is unit-weight composite reliability attributable to g. PCA variance is a separate exploratory statistic. Fit indices are approximate for bounded aggregate scores and this small, selected sample.

Item bootstrap draws common question IDs jointly across models within each task. Family/item bootstrap additionally samples curated family clusters. Scores are standardized over the original panel. Group assignments stay fixed for intervals and are rediscovered separately. Item-only variance estimates shift downward: percentile ranges are nominal, uncalibrated sensitivity summaries, not coverage guarantees. Bias diagnostics are exported.

Seed 0; 1000 replicates per interval mode; 200 rediscovery replicates. Nonconvergence, singularity and boundary fits are recorded. Nominal intervals are withheld below 90% admissible replicates. No silent matrix regularization is used.

Family provenance is in families.csv. Documented series are a clustering assumption, not proof of shared weights; unresolved aliases remain singleton clusters. Relations between clusters may remain. Historical generation settings, provider differences and repeated-run uncertainty are not fully identified.

The broad usable archive covers only part of 2024. The latest HF judgment pool has three domains, and the newer export repository returned 404. Independent reproduction of published leaderboard totals remains pending. Data-card licensing is unspecified; this report does not clear raw-data redistribution.

Sources: https://huggingface.co/livebench ; https://github.com/LiveBench/LiveBench . Per-file immutable URLs/checksums, code hashes and package versions are in metadata.json. Exact family-source links are in families.csv.